

Assignment 04: Agentic PDF Generation Framework

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1 Introduction

This document outlines the architecture for Assignment 04, focusing on autonomous document generation. By integrating LaTeX with agentic workflows, we aim to bridge the gap between structured technical writing and intelligent automation.

2 Assignment Context

The objective is to establish an end-to-end pipeline capable of generating academic-grade PDFs. This requires not only typesetting capabilities but also logical control over document structure and content verification.

3 Technical Foundation

3.1 LuaLaTeX and CLS Templates

We leverage LuaLaTeX for its advanced scripting capabilities, allowing for programmatic manipulation of the document model. The project utilizes a custom `custom-academic-template.cls` to ensure stylistic consistency and adherence to academic formatting standards [1].

3.2 Agent Orchestration and Skills

Content generation is mediated by an orchestrator agent that delegates specialized tasks to defined “skills”. Skills act as modular agents, ensuring clean separation of concerns between content creation, compilation, and quality assurance.

4 Methodology

4.1 QA Workflow

Our QA process employs a multi-faceted approach, checking document structure, citation integrity, and language quality. This ensures that the generated output meets the rigors of formal documentation.

5 Repository Structure

The project follows a modular organization:

- `src/`: Core logic (orchestrator, compilation, QA).
- `skills/`: Declarative agent capabilities.
- `assets/`: External resources (images, graphs).

Table 1: Project Components

Component	Description
Orchestrator	Manages the workflow
LaTeX Builder	Compiles the document
QA Suite	Ensures quality and consistency

6 Results

7 Visualizations

Figure Placeholder

Figure 1: Placeholder for an asset/image.

8 Reproducibility, Limitations, and Future Work

While the current implementation provides a robust skeleton, reproducibility relies on specific environment configurations. Limitations currently include manual intervention for complex layout adjustments. Future improvements will focus on fully automated asset generation and dynamic bibliography management.

9 Conclusion

This assignment demonstrates the viability of agent-based document generation [1]. The proposed framework offers a scalable path for automating high-quality academic publications.

References

- [1] F. Mittelbach and M. Goossens, *The LaTeX Companion*. Addison-Wesley, 2004.